Super resolution at the CMS detector using Generative Adversarial Networks

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1 Introduction

In high-energy physics (HEP) experiments, such as those conducted at particle colliders, collisions between particles produce collimated sprays of secondary particles known as *jets*. These jets arise from the hadronization of quarks and gluons and serve as important observables for studying fundamental interactions.

Due to limitations in detector resolution, the recorded data is often available at a relatively low resolution. This can obscure fine-grained structures within jets that are crucial for tasks such as particle identification, anomaly detection, and precision measurements.

The problem of reconstructing high-resolution (HR) images from low-resolution (LR) inputs is therefore of significant importance. Improving resolution can enhance the interpretability of jet substructure and potentially improve the performance of downstream machine learning models used in physics analyses.

In this task, we address the problem of **jet image super-resolution** using deep learning techniques. Specifically, we formulate the task as an image-to-image translation problem and employ Generative Adversarial Networks (GANs) to learn a mapping from low-resolution calorimeter images to their high-resolution counterparts.

2 Understanding of the Dataset

The dataset used in this work consists of simulated particle jets interacting with a calorimeter detector. Simulation provides access to both low-resolution and corresponding high-resolution representations, making it suitable for supervised learning.

Each sample in the dataset contains:

- Low-resolution image: $3 \times 64 \times 64$
- High-resolution image: $3 \times 125 \times 125$
- Transverse momentum (p_T)
- Jet mass (m_0)
- Jet label (quark or gluon)

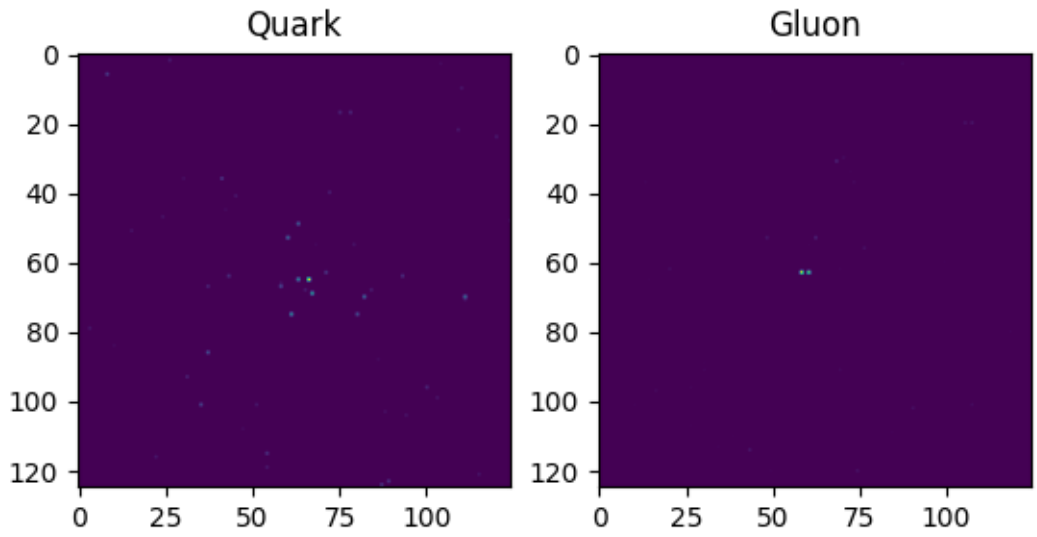


Figure 1: Quark and Gluon Visualization

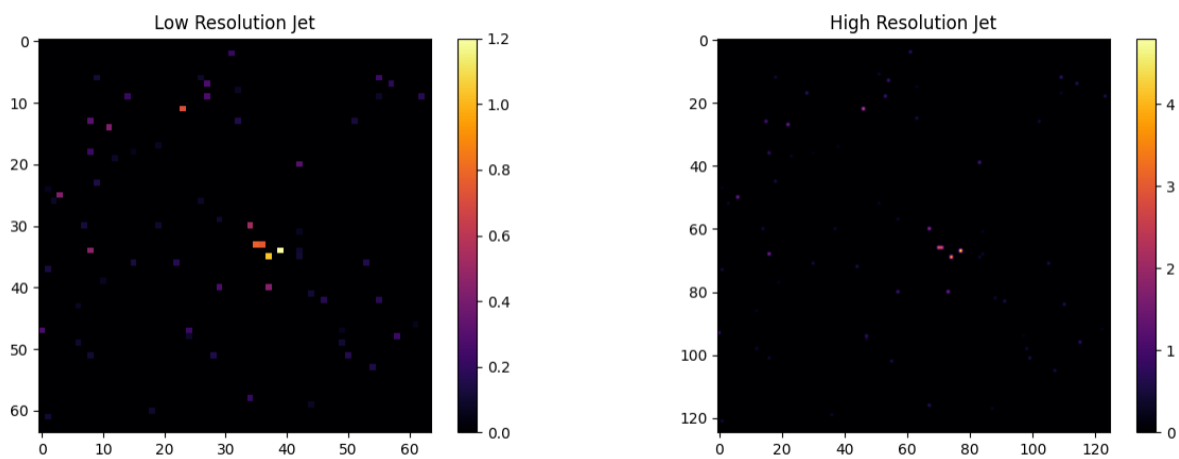


Figure 2: Low resolution and High Resolution Data Visualization

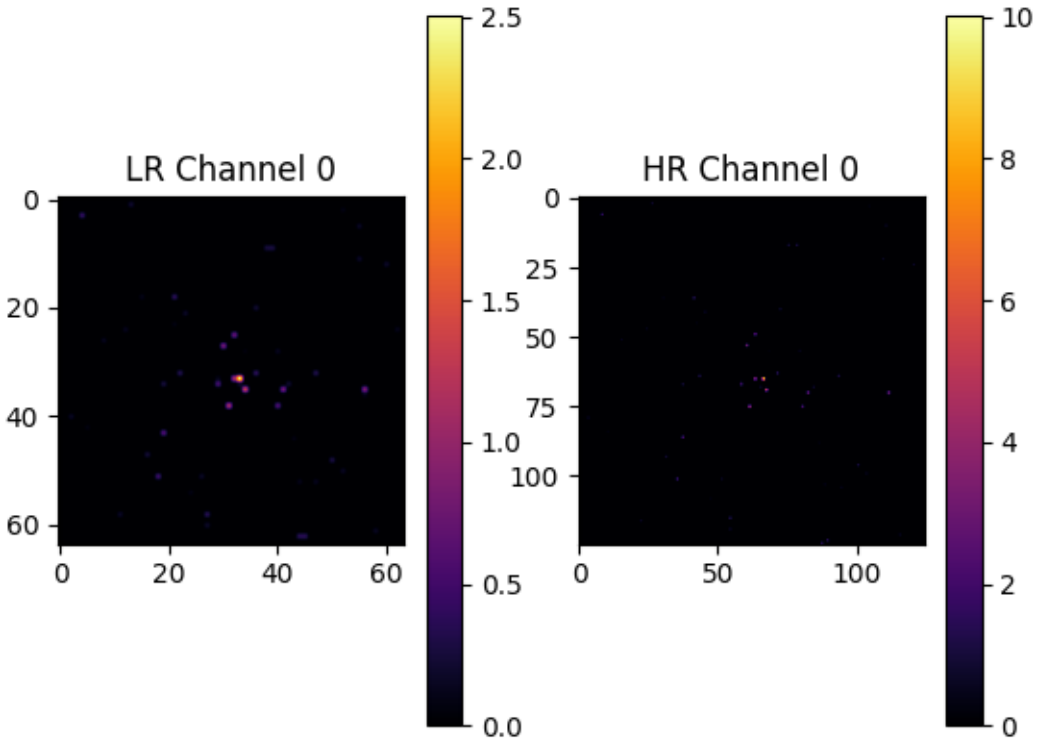


Figure 3: Low resolution(Channel 0) and High Resolution(Channel 0) Visualization

The three channels represent different physical measurements (e.g. electromagnetic energy, hadronic energy, or other detector-specific signals), analogous to RGB channels in standard image processing tasks.

A defining characteristic of this dataset is its **extreme sparsity**. Approximately 98% of the pixels contain zero values, reflecting the fact that energy depositions are highly localized along particle trajectories. This sparsity poses a significant challenge for learning algorithms, as useful information is concentrated in a very small subset of pixels.

3 Data Preprocessing

The dataset was provided in `.parquet` format, which is efficient for storage and data transfer. However, direct loading from .parquet during training introduces significant overhead due to repeated disk I/O operations.

To improve training efficiency, data was converted into the PyTorch Tensor Format (.pt), enabling faster loading and seamless integration with the training pipeline.

3.1 Chunking Strategy

Due to the large size of the dataset (several gigabytes), it was not feasible to load all samples into my device memory simultaneously. To address this, the dataset was divided into smaller chunks of approximately 10,000 samples each.

This chunking approach provides several advantages:

- Reduces peak memory usage
- Enables training on CPU-only systems
- Allows scalable processing of large datasets

Chunks were loaded sequentially during training, ensuring that the model could eventually see the entire dataset without exceeding memory limits.

3.2 Normalization

Each sample was normalized independently using:

$$\mathrm{x}_{\text{normalized}} = \frac{x}{\max(x)+\epsilon}$$

where ϵ is a small constant added to avoid division by zero.

This normalization ensures that all inputs lie within a comparable numerical range, improving training stability. It also prevents high-energy outliers from dominating the learning process and allows the model to focus on relative spatial patterns rather than absolute magnitudes.

3.3 Data Representation

The jet images were treated as three-channel tensors, analogous to color images. This representation enables the use of convolutional neural networks (CNNs), which are well-suited for capturing spatial correlations and hierarchical features.

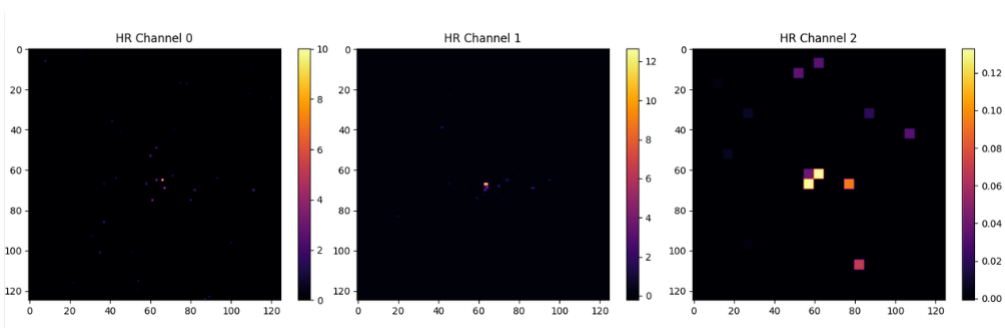


Figure 4: The three channel representations of one sample of a high-resolution image

4 Problem Formulation

The goal of this project is to learn a function:

[G: $\mathrm{X}_{\mathrm{LR}} \rightarrow X_{\mathrm{HR}}$]

where G is a neural network that maps low-resolution inputs to high-resolution outputs.

This is a **conditional generation problem**, as the output is explicitly conditioned on the input image. Unlike unconditional generative models, the network must preserve the underlying structure of the input while enhancing its resolution.

The task can also be viewed as a structured prediction problem, where spatial consistency and physical plausibility are critical.

5 Model Architecture

We employ a conditional GAN framework inspired by the Pix2Pix architecture, which has proven effective for image-to-image translation tasks. The model consists of two components: a generator and a discriminator, trained adversarially to produce realistic high-resolution jet images from low-resolution inputs.

5.1 Generator: U-Net Architecture

The generator is based on a U-Net architecture, consisting of:

- An encoder that progressively downsamples the input image to extract high-level features and consists of convolutional layers that progressively downsample the input image, extracting hierarchical features.
- A decoder that upsamples these features to reconstruct the high-resolution output and uses transposed convolutions
- Skip connections that link corresponding encoder and decoder layers

The skip connections play a crucial role by:

- Preserving fine-grained spatial information
- Enabling direct propagation of low-level features
- Preventing loss of detail during downsampling

This architecture is particularly well-suited for super-resolution tasks, where both global structure and local detail must be preserved.

5.2 Discriminator: PatchGAN

The discriminator is implemented as a PatchGAN, which operates on local patches rather than the entire image.

Key characteristics include:

- Classifies each patch as real or fake
- Focuses on high-frequency details
- Encourages sharper and more realistic outputs

The discriminator takes as input the concatenation of the LR image and either the real or generated HR image, ensuring that it evaluates consistency between input and output. It consists of a series of convolutional layers with increasing feature depth and LeakyReLU activations. These layers progressively reduce spatial dimensions while learning to distinguish real from generated data at the patch level. The output is a feature map representing the probability of each patch being real or fake, rather than a single scalar value.

5.3 Architecture Summary

Component	Layer	Description
Generator	Enc1	Conv2D ($3 \to 64$, stride 2)
Generator	Enc2	Conv2D ($64 \to 128$, stride 2)
Generator	Enc3	Conv2D ($128 \to 256$, stride 2)
Generator	Dec1	Transposed Conv ($256 \to 128$)
Generator	Dec2	Transposed Conv ($128 \to 64$)
Generator	Output	Conv2D ($64 \to 3$)
Discriminator	Input	Concatenated LR + HR (6 channels)
Discriminator	Conv1	Conv2D ($6 \to 64$)
Discriminator	Conv2	Conv2D ($64 \to 128$)
Discriminator	Conv3	Conv2D ($128 \to 256$)
Discriminator	Output	Conv2D ($256 \to 1$, patch map)

Table 1: Summary of generator and discriminator architectures

6 Loss Functions

The generator is trained using a combination of adversarial and reconstruction losses:
[$\mathrm{L}_G = \mathcal{L}_{adv} + \lambda \mathcal{L}_{L1}$]
where:

- $\mathcal{L}_{adv}$ encourages the generator to produce outputs indistinguishable from real images
- $\mathcal{L}_{L1}$ enforces pixel-wise similarity between generated and ground truth images
- $\lambda = 100$ balances perceptual realism and structural accuracy
- The L1 loss is preferred over L2 loss as it reduces excessive blurring and preserves sharper structures.
- The discriminator is trained using binary cross-entropy loss to distinguish real from generated samples.
- Here the adversarial loss is also taken as binary cross-entropy loss which quantifies how well generator fools discriminator.

7 Training Strategy

7.1 Optimization

The model was trained using the Adam optimizer with the following hyperparameters:

- Generator learning rate: 2×10^{-4}
- Discriminator learning rate: 1×10^{-4}

7.2 Stabilization Techniques

Training GANs is known to be unstable. To mitigate this:

- The generator was updated more frequently than the discriminator
- The discriminator learning rate was reduced to prevent overpowering
- Visual monitoring was used to track qualitative progress

7.3 Hardware Constraints

Due to CPU-only training:

- A small batch size (8) was used
- Training was performed in chunks
- The number of epochs was limited

These constraints increased training time but did not prevent meaningful learning.

8 Challenges Encountered

8.1 Sparse Data

Extreme sparsity led to:

- Weak gradient signals
- Difficulty learning low-energy regions

8.2 GAN Instability

Observed issues included:

- Rapid discriminator convergence
- Oscillating generator loss

9 Experimental Comparison

To understand the effect of training duration and dataset size, three models were trained under different configurations.

9.1 Model Configurations

- **Model A:** Trained on 1 data chunk for 12 epochs
- **Model B:** Trained on 1 data chunk for 20 epochs (Architecture changes below)
  - UNet-style Generator with skip connections (fixes structure learning)
  - Removed tanh (prevents output collapse)
  - Weaker Discriminator (lower LR)
  - Train Generator twice per step (balance training)
  - Per-sample normalization (better for sparse data)
- **Model C:** Trained on 5 data chunks for 10 epochs (Same Architecture as Model B)

9.2 Training Progression

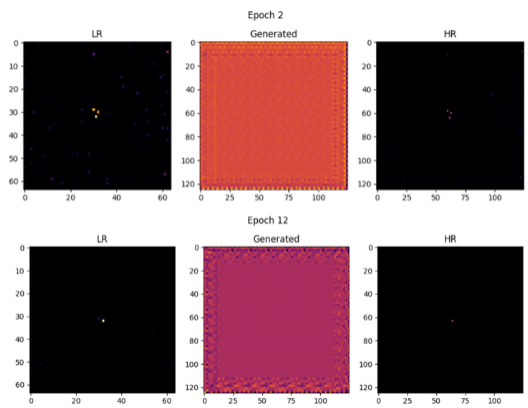


Figure 5: Model A - Evolution of generated images across epochs: The generated outputs lack clear structure and fail to capture the central jet region. This indicates insufficient training and that some architecture change is needed.

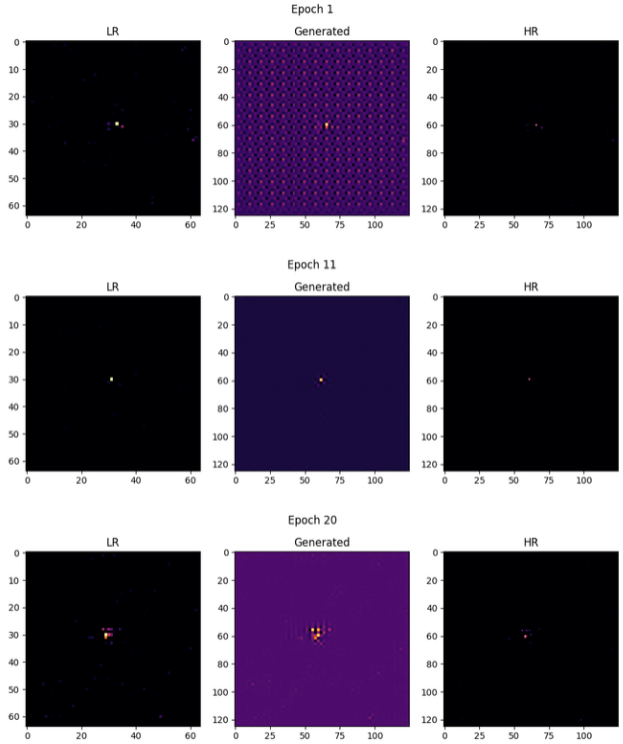


Figure 6: Model B - Evolution of generated images across epochs: The model initially learns the central jet structure but becomes unstable in later epochs. The discriminator dominates, leading to degraded outputs.

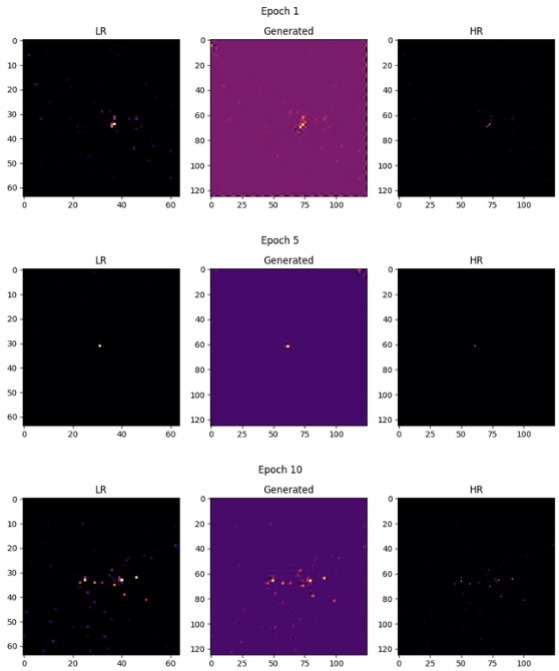
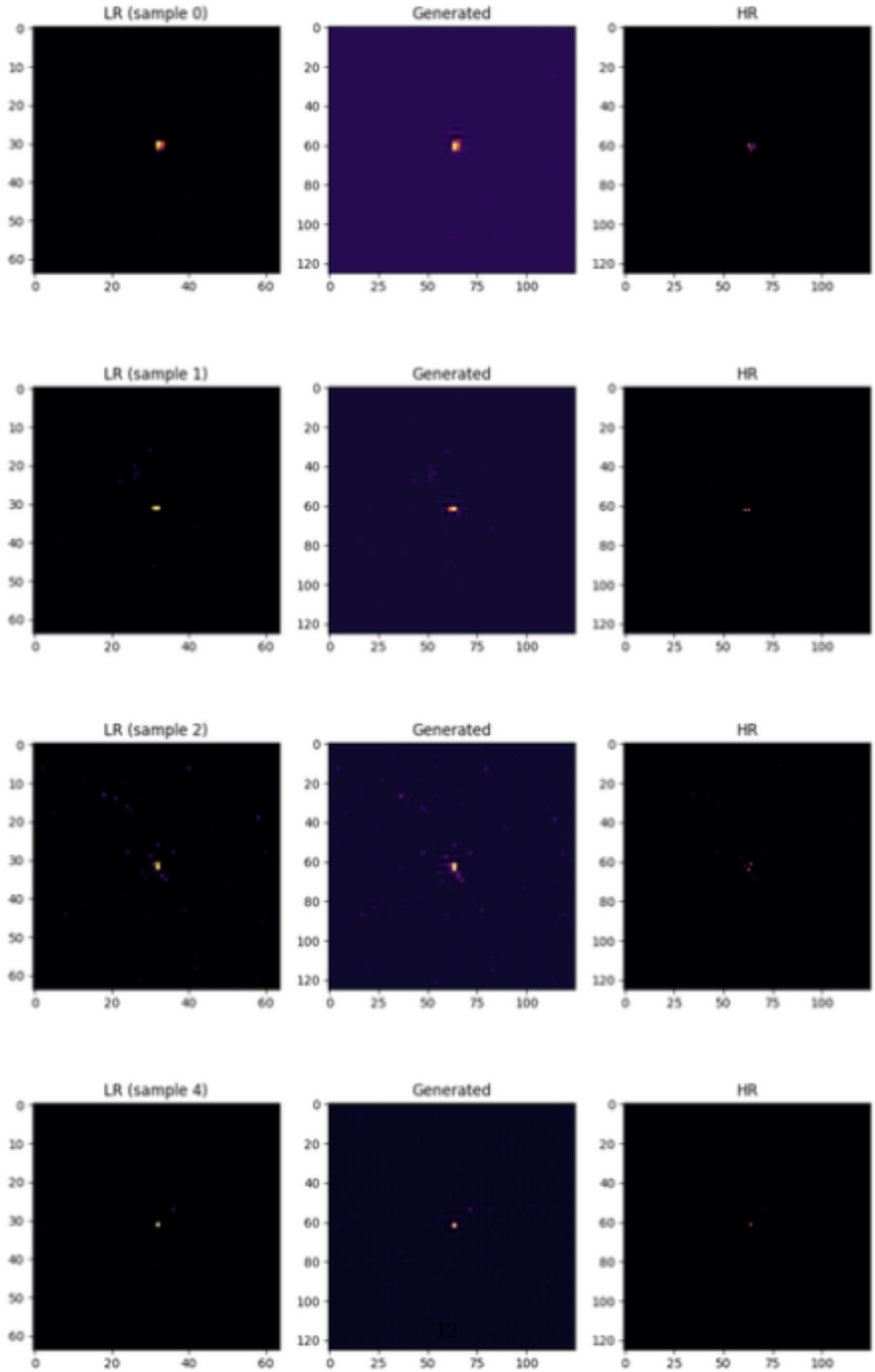


Figure 7: Model C - Evolution of generated images across epochs: Using more data improves generalization slightly, but the model still struggles to reproduce fine details. Outputs capture coarse structure but remain smooth and partially noisy.

9.3 Sample Reconstructions



9.4 Observations

- **Model A:** The model is undertrained and fails to reconstruct meaningful jet structures. The generated outputs lack clear structure and fail to capture the central jet region. This indicates insufficient training and that some architecture change is needed.
- **Model B:** While the model learns the central structure, training instability leads to degraded outputs and increased noise in later epochs. This reflects typical GAN behavior where the discriminator dominates.
- **Model C:** This configuration achieves the best performance. The model successfully reconstructs the central jet structure while maintaining stability across training. The use of multiple data chunks improves generalization, but the model still struggles to reproduce fine details. Outputs capture coarse structure but remain smooth and partially noisy.

9.5 Training Dynamics

The training process was monitored using generator loss (G_{loss}) and discriminator loss (D_{loss}).

During early epochs, both losses remained relatively balanced, indicating stable adversarial learning. However, as training progressed, the discriminator loss frequently dropped to near-zero values while the generator loss increased significantly.

This behavior suggests that the discriminator became overly dominant, making it difficult for the generator to improve. Such instability is a well-known challenge in GAN training and was observed consistently across experiments.

Representative training logs are shown below:

- Early training: $G_{loss} \approx 1$, $D_{loss} \approx 1$
- Later training: $G_{loss} \gg 10$, $D_{loss} \to 0$

These trends indicate that the model enters an unstable regime in later epochs, limiting further improvements in output quality.

9.6 Discussion

These results highlight the importance of balancing training duration and dataset size in GAN-based models. Training for too few epochs leads to underfitting, while excessive training can result in instability due to adversarial dynamics.

Using a moderate number of epochs with a sufficiently diverse dataset yields the best results, as it allows the model to learn meaningful structure without diverging. Yet the model does not fully reconstruct high-resolution details.

10 Energy Correlation Analysis

To evaluate whether the generated images preserve physically meaningful information, we compare the total energy of the reconstructed jets with the ground truth.

For each sample, the total energy is computed as:

$$E = \sum_{c,h,w} x_{c,h,w}$$

We then plot the correlation between the real high-resolution energy and the generated energy.

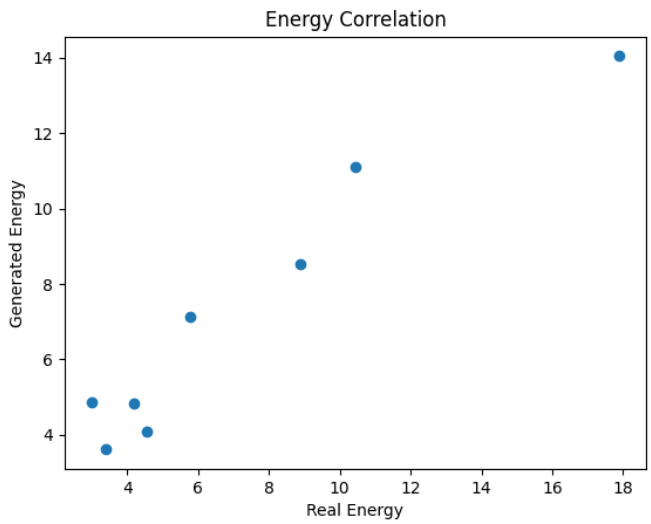


Figure 9: Correlation between real and generated jet energies

The results show positive correlation between real and generated energies which indicates that the model preserves global physical properties of the jet, even if local details are not perfectly reconstructed.

From the results, we observe that the generated images exhibit a clear positive correlation with the ground truth energy. This suggests that the model is able to capture the overall energy scale of the jets, which is a key physical characteristic.

However, deviations from the ideal diagonal line indicate that the model does not perfectly reproduce the exact energy distribution. These discrepancies can be attributed to smoothing effects introduced by the reconstruction loss and the inherent difficulty of modeling sparse high-dimensional data.

Overall, this analysis demonstrates that the model not only produces visually plausible outputs but also retains meaningful physical information.

11 Discussion

The model successfully reconstructs the core structure of particle jets and captures major energy deposition regions.

Key observations:

- Accurate localization of central jet energy
- Preservation of global structure
- Loss of fine details
- Generated image does not match the resolution of the high-resolution image

These results demonstrate that GANs can learn meaningful representations even in highly sparse domains.

12 Conclusion

We presented a GAN-based approach for super-resolution of jet images. Despite challenges related to sparsity, instability, and hardware limitations, the model achieved promising results.

Future improvements could include:

- Diffusion-based generative models for better stability
- Transformer architectures for capturing long-range dependencies
- Physics-informed loss functions to enforce conservation laws